

WundCheck — Submission package

FHNW MSc Medical Informatics • Clinical Decision Support Systems 2026 • 4 August 2026

A mini knowledge-based CDSS for the daily self-check of post-operative wounds, developed along the WHO SMART Guidelines layer model (L1 → L2 → L3 → L4).

What the assignment asks for, and where it is

Assignment requirement	File
Presentation materials	WundCheck_CDSS_Presentation.pptx · .pdf (15 slides + 10 backup slides, with speaker notes) · speaking script: 14-presentation-script.pdf
L2 — high-level workflow (BPMN)	BPMN_workflow-highlevel.bpmn / .svg / .png · documented in 04-l2-workflow-bpmn.pdf
L2 — data dictionary	05-l2-data-dictionary.pdf — 48 elements
L2 — decision logic	06-l2-decision-logic.pdf — 18 rules · additionally as a BPMN gateway cascade in BPMN_workflow-app-detail.bpmn
L3 documentation	08-l3-architecture.pdf · machine-readable file: L3_wundcheck-l3.json
Pre-recorded demo	WundCheck_L4_Demo.mp4 — 1:46, 1920×1080, a screen recording of the running application: the full patient journey plus the threshold change
Access to the L4	L4_wundcheck-app.html — open in any browser, no installation, works offline

Everything else

#	Document	Assignment item
01	01-health-need-and-scope.pdf	Health / health system need
02	02-personas-and-scenarios.pdf	User personas and scenarios
03	03-l1-guidelines-and-evidence.pdf	Guidelines and evidence
04	04-l2-workflow-bpmn.pdf	L2 — high-level workflow
05	05-l2-data-dictionary.pdf	L2 — data dictionary
06	06-l2-decision-logic.pdf	L2 — decision logic
07	07-l2-l1-to-l2-challenges.pdf	L1 → L2 translation challenges
08	08-l3-architecture.pdf	L3 — machine-readable code
09	09-l4-implementation-ux.pdf	L4 — executable layer and UI considerations
10	10-qa-testing.pdf	Quality assurance testing
11	11-implementation-monitoring-evaluation.pdf	Implementation, monitoring and evaluation
12	12-impact-and-regulation.pdf	Expected impact (and regulatory classification)
13	13-project-plan-and-contributions.pdf	Project plan, presentation outline, contributions
14	14-presentation-script.pdf	Speaking script for all 25 slides — timings, handovers, Q&A jump targets

#	Document	Assignment item
—	README.pdf	Repository overview
—	example-fhir-bundle.json	Example FHIR R4 transaction bundle produced by the L4

Running the L4

Open `L4_wundcheck-app.html` in any modern browser. No server, no installation, no dependencies; it works fully offline. The application is bilingual (English by default, German switchable in the header).

The **L3 inspector** at the foot of the page shows the loaded L3 version, the derived values, the FHIR endpoint and the outbound queue — and it lets you edit the thresholds live. Changing `fever_orange` from 38.0 to 37.5 and re-evaluating the same case changes the classification without a rebuild. That is the clearest demonstration of the layer separation.

Load `L3 file` accepts any other L3 JSON; the app renders whatever it is given.

Key figures

Data elements	48 (9 derived · 38 with a terminology binding · 47 with a FHIR mapping)
Decision rules	18 (13 exclusive, worst-first · 5 non-exclusive information rules)
Documented L1 → L2 challenges	9
Hazards identified and ranked	6
Automated checks	19 decision-logic tests · 20 FHIR R4 conformance checks · 1 no-hardcoding check — all passing
Defects found and fixed	8

Two things worth knowing before reading

Three of the eight defects were invisible to a document review. B-07 and B-08 only surfaced once the rules were actually executed: a trend rule referenced its own result and could never fire, and after that was fixed the rule turned out to be unreachable against the worst-first cascade. B-06 was a contradiction between two of our own documents. All three are documented with root cause, fix and retest in `10-qa-testing.pdf`.

The claim “the L4 contains no clinical content” is machine-enforced. A check scans the application source for every element id, section id, rule id, threshold, code system URI and the FHIR endpoint, and fails if any of them appears. It found two real leaks on its first run, both of which were fixed by moving the content into the L3.

Reproducing the checks

From the repository root:

```
node tools/run-tests.mjs      # 19 decision-logic test cases
node tools/check-fhir.mjs     # 20 FHIR R4 conformance checks
node tools/check-hardcoding.mjs # proves the L4 holds no clinical content
```

```
python3 tools/embed-l3.py          # re-embed the L3 into the app after an L3 change
python3 tools/build-submission.py  # rebuild this package
node    tools/build-presentation.js # rebuild the slide deck
python3 tools/build-demo-video.py  # rebuild the demo video
```

 **Study prototype — not for clinical use.** WundCheck triages and documents; it does not replace medical assessment.